

Roham Koohestani

Researcher, JetBrains Research (HAX Team)
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Research Interests

AI for software engineering: large language models for code, developer–AI interaction, and the verification of agentic systems. My aim is AI assistance that developers can trust, grounded in empirical evidence of what they need and in formal guarantees designed to serve those needs.

Employment

JetBrains Research The Hague, Netherlands 2026 – present
Researcher, Human-AI Experience (HAX) team. Developer-centric trustworthy AI for software engineering.

JetBrains Research Amsterdam, Netherlands 2025 – 2026
Research Intern, Human-AI Experience team. In-IDE calibration of LLMs at production scale (JetBrains–TU Delft AI4SE collaboration).

AISE Lab, Delft University of Technology Delft, Netherlands 2024 – present
Research Assistant, supervised by Prof. Maliheh Izadi. Benchmarking, agent verification, and IDE personalization.

Syntho B.V. Amsterdam, Netherlands 2023 – 2024
Software Engineering Intern. Synthetic data generation, benchmarking, and time-series synthesis conditioned on metadata.

Syntho B.V. Amsterdam, Netherlands 2022 – 2023
High School Intern. Bias minimization and fairness in machine learning through synthetic data.

Education

MSc Computer Science Delft University of Technology 2026 – present
Thesis: AI-agent supported migration of legacy systems at the Dutch Inspectorate for Living Environment and Transport (ILT).
Supervisor: Prof. Arie van Deursen.

Mathematics Minor University of Amsterdam 2025 – present
Number and set theory, theory of statistics, differential equations, information theory, and Markov chains.

BSc Computer Science and Engineering Delft University of Technology 2023 – 2026
Data elective track. Bachelor thesis: AST-guided membership inference for code language models (published at FSE Companion 2026).

Publications

*Author ordering follows contribution. **Bold** indicates the presenting / corresponding author where*

applicable. Underline indicates Roham Koohestani.

Under Review

- [A2] Roham Koohestani, Ziyu Li, Anton Podkopaev, Maliheh Izadi. *Memory-Constrained Probabilistic Trace Models for Agentic AI*. Formal Aspects of Computing (FAC). *Under review*. [arXiv](#)
- [A1] Roham Koohestani, Ateş Görpelioglu, Egor Klimov, Burcu Kulahcioglu Ozkan, Maliheh Izadi. *TriCEGAR: A Trace-Driven Abstraction Mechanism for Agentic AI*. Preprint; co-first author. [arXiv](#) / [artifact](#)

Journal Publications

- [J1] Roham Koohestani, Philippe de Bekker, Begüm Koç, Maliheh Izadi. *Benchmarking AI Models in Software Engineering: A Review, Search Tool, and Enhancement Protocol*. IEEE Transactions on Software Engineering (TSE), 2025. Journal-first presentation at FSE 2026. [arXiv](#) / [code](#) / [BenchScout](#)

Conference Publications

- [C5] Roham Koohestani, Agnia Sergeyuk, David Grosse, Claudio Spiess, Sergey Titov, Prem Devanbu, Maliheh Izadi. *Does In-IDE Calibration of Large Language Models Work at Scale?* FSE Companion 2026, Industry Track. Co-first author. [DOI](#) / [arXiv](#)
- [C4] Roham Koohestani, Ali Al-Kaswan, Jonathan Katzy, Maliheh Izadi. *AST-PAC: AST-guided Membership Inference for Code*. FSE Companion 2026, IVR Track. From bachelor thesis. [DOI](#) / [artifact](#)
- [C3] Roham Koohestani, Parham Bateni, Aydin Ebrahimi, Behdad Etezadi, Kiarash Karimi, Maliheh Izadi. *Code4Me V2: A Research-oriented Code-completion Platform*. ICSE Companion 2026, Demonstrations Track. Co-first author. [DOI](#) / [arXiv](#)
- [C2] Amirhossein Deljouyi, Roham Koohestani, Maliheh Izadi, Andy Zaidman. *Leveraging Large Language Models for Enhancing the Understandability of Generated Unit Tests*. ICSE 2025. [arXiv](#) / [code](#)
- [C1] Roham Koohestani, Maliheh Izadi. *HyperSeq: A Hyper-Adaptive Representation for Predictive Sequencing of States*. FSE Companion 2025. [DOI](#) / [code](#)

Workshop Publications

- [W2] Roham Koohestani. *AgentGuard: Runtime Verification of AI Agents*. ASE 2025 Workshops (SE Agents). Sole author. [DOI](#) / [arXiv](#)
- [W1] Roham Koohestani, Maliheh Izadi. *Rethinking IDE Customization for Enhanced HAX: A Hyperdimensional Perspective*. IDE Workshop 2025 (co-located with ICSE). [DOI](#)

Teaching

Teaching Assistant Delft University of Technology 2024 – 2026
CS4570 Machine Learning for Software Engineering.

Supervised 50+ MSc students in research projects over two academic years, across a 10-week intensive group-research course. Coordinated assessment for a cohort of more than 130 students.

Service

Co-Reviewing — Conferences & Journals

ICSE ('25, '26C1, '26C2), FSE ('25), ASE ('25), EMSE ('25), ISSTA ('26)

Reviewer — Conference

ICSME Industry Track 2026

Reviewer — Workshops

DeepLearning4Code Workshop @ NeurIPS 2025

DeepLearning4Code Workshop @ ICML 2026

FSE × AIWARE Competition — Poisoned Chalice @ FSE 2026

Organizing

Web-Chair, AgenticSE Workshop @ ASE 2025